Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
MERN Practical

Practical – IV

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Aim: React Components, Props, and State.

4.A). Create multiple components and render them on a single page.

SOURCE CODE:-

App.JSX

```
import React from 'react';
import Header from './components/Header';
import Footer from './components/Footer';
import Student from './components/Student';
import Result from './components/Result';
import Attendance from './components/Attendance';
import './App.css';
const App = () => {
  return (
    <div className="App">
      <div className="container">
        <Header />
        <Student />
        <Attendance />
        <Result />
        <Footer />
      </div>
    </div>
  );
}
export default App;
```

Header.jsx

```
import React from 'react';
const Header = () => {
  return (
    <div className="header">
      <h1>🎓 M. V. L. U. College Placement
Dashboard</h1>
      <div className="divider"></div>
    </div>
  );
};
export default Header;
```

Student.jsx

```
import React from 'react';
const Student = () => {
  // Student data
  const studentData = {
    name: 'Lavkush Vishwakarma',
    rollNo: '48',
    branch: 'Information Technology',
    semester: 'VI',
    cgpa: '9.99'
  };
  return (
    <div className="card">
      <h2 className="heading">Student Details</h2>
      <div className="info">
        <div className="item">
          <span className="label">Name :</span>
          <span
className="value">{studentData.name}</span>
        </div>
        <div className="item">
          <span className="label">Roll No :</span>
          <span
className="value">{studentData.rollNo}</span>
        </div>
        <div className="item">
          <span className="label">Branch :</span>
          <span
className="value">{studentData.branch}</span>
        </div>
        <div className="item">
          <span className="label">Semester :</span>
          <span
className="value">{studentData.semester}</span>
        </div>
        <div className="item">
          <span className="label">CGPA :</span>
          <span
className="value">{studentData.cgpa}</span>
```

```
      </div>
    </div>
  </div>
 );
};
export default Student;
```

Attendance.jsx

```
import React from 'react';
const Attendance = () => {
 // Attendance data
 const attendance = {
  percentage: 88,
  minimum: 75
 };

 // Calculate eligibility
 const isEligible = attendance.percentage >=
attendance.minimum;
return (
  <div className="card">
   <h2 className="heading">Attendance</h2>
   <div className="info">
    <div className="item">
     <span className="label">Attendance :</span>
     <span
className="value">{attendance.percentage}%</span>
    </div>
    <div className="item">
     <span className="label">Minimum Attendance
:</span>
     <span
className="value">{attendance.minimum}%</span>
    </div>
    <div className="item">
     <span className="label">Status :</span>
     <span className={isEligible ? 'status-eligible' :
'status-not-eligible'}>
       {isEligible ? 'Eligible' : 'Not Eligible'}
     </span>
    </div>
   </div>
  </div>
 );
};
export default Attendance;
```

Result.jsx

```
import React from 'react';
const Result = () => {
 // Academic result data
 const totalMarks = 520;
 const maxMarks = 600;
 const percentage = ((totalMarks / maxMarks) *
100).toFixed(2);
 // Calculate grade based on percentage
 let grade = 'F';
 const percentNum = parseFloat(percentage);
 if (percentNum >= 80) grade = 'A';
 else if (percentNum >= 70) grade = 'B';
 else if (percentNum >= 60) grade = 'C';
 else if (percentNum >= 50) grade = 'D';
 return (
  <div className="card">
   <h2 className="heading">Academic Result</h2>
   <div className="info">
    <div className="item">
     <span className="label">Total Marks :</span>
     <span className="value">{totalMarks}</span>
    </div>
    <div className="item">
     <span className="label">Maximum Marks
:</span>
     <span className="value">{maxMarks}</span>
    </div>
    <div className="item">
     <span className="label">Percentage :</span>
     <span className="value">{percentage}%</span>
    </div>
    <div className="item">
     <span className="label">Grade :</span>
     <span className={`grade-
${grade}`}>{grade}</span>
    </div>
   </div>
  </div>
 );
};
export default Result;
```

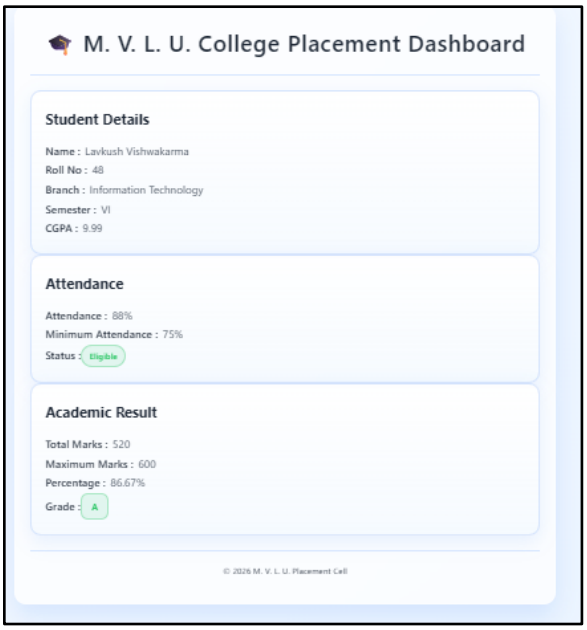
Footer.jsx

```
import React from 'react';
const Footer = () => {
 return (
  <div className="footer">
   <p>© 2026 M. V. L. U. Placement Cell</p>
  </div>
 );
};
export default Footer;
```

main.jsx

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'
createRoot(document.getElementById('root')).render(
 <StrictMode>
  <App />
 </StrictMode>,
)
```

OUTPUT:-



4.B). Pass data between components using props.

SOURCE CODE:-

Header.jsx:-

```
function Header(){
    return(
        <div>
            <h1> Lavkush Online food delivery
            Dashboard </h1>
            <hr/>
        </div>
    );
}
export default Header;
```

Footer.jsx:-

```
function Footer(){
    return(
        <div>
            <h3>@ 2026 Food Delivery App</h3>
        </div>
    );
}
export default Footer;
```

HotelCard.jsx:-

```
function HotelCard(props) {
  const discount = (props.price * props.discount) / 100;
  const finalPrice = props.price - discount;
  return (
    <div>
      <h2>{props.restaurant}</h2>
      <p>Food Item :{props.food}</p>
      <p>Original Price:${props.price}</p> <p>Discount
:{props.discount}%</p>{" "}
      <p>Discount Amount :${discount}</p>{" "}
      <p>
        <b>Final Price: ${finalPrice}</b>
      </p>{" "}
      <p>Ratings :{props.rating}</p>
    </div>
  );
}
export default HotelCard;
```

OUTPUT:-



App.jsx:-

```
import Header from "./components/Header";
import Footer from "./components/Footer";
import HotelCard from "./components/HotelCard";
function App() {
  return (
```

```
    <div>
      <Header />
      <HotelCard
        restaurant="Mojo Pizza"
        food="Farmhouse PIZZA"
        price={499}
        discount={45}
        rating={7.8}
      />
      <HotelCard
        restaurant="Mojo Pizza"
        food="Paneer PIZZA"
        price={699}
        discount={45}
        rating={8.8}
      />
      <HotelCard
        restaurant="Mojo Pizza"
        food="Paneer Farmhouse PIZZA"
        price={799}
        discount={45}
        rating={9.8}
      />
    </div>
  );
}
export default App;
```

4.C). Use the useState hook to manage and update data.

SOURCE CODE:-

```
import React, { useState } from 'react';
const Counter = () => {
  const [count, setCount] = useState(0);
  const buttonStyle = {
    padding: '6px 12px',
    backgroundColor: '#007bff',
    color: '#fff',
    border: 'none',
    borderRadius: '4px',
    cursor: 'pointer',
    margin: '0 4px',
  };
  return (
    <div className="card">
      <h2>Counter</h2>
      <div className="info">
        <span className="value">{count}</span>
        <div style={{ display: 'flex', gap: '8px', marginTop:
'12px' }}>
          <button onClick={() => setCount(count + 1)}
style={buttonStyle}>
            +
          </button>
          <button onClick={() => setCount(count - 1)}
style={buttonStyle}>
            -
          </button>
          <button onClick={() => setCount(0)}
style={buttonStyle}>
            Reset
          </button>
        </div>
      </div>
    </div>
  );
};
export default Counter;
```

OUTPUT:-



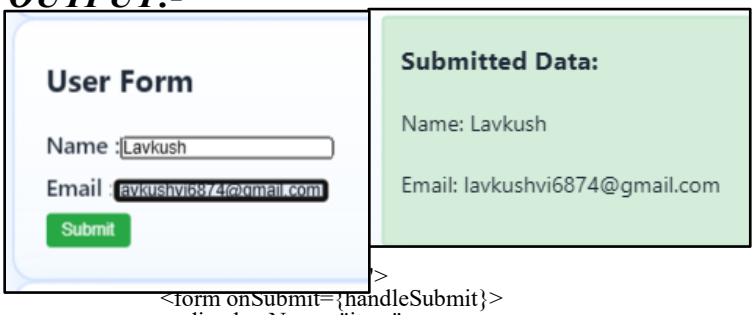
4.D). Create a form component to accept user input and display it dynamically.

SOURCE CODE:-

UserForm.jsx:-

```
import React, { useState } from 'react';
const UserForm = ({ onAddUser }) => {
  const [name, setName] = useState('');
  const [email, setEmail] = useState('');
  const [submitted, setSubmitted] = useState(null);
  const handleSubmit = (e) => {
    e.preventDefault();
    if (!name.trim() || !email.trim()) return;
    const newUser = { id: Date.now(), name: name.trim(),
email: email.trim() };
    onAddUser(newUser);
    setSubmitted(newUser);
    setName('');
    setEmail('');
  };
  return (
    <div className="card">
      <h2>User Form</h2>
      <form onSubmit={handleSubmit}>
        <div className="item">
          <span className="label">Name :</span>
          <input
            type="text"
            value={name}
            onChange={(e) => setName(e.target.value)}
            placeholder="Enter name"
```

OUTPUT:-



```
          />
        </div>
        <div className="item">
          <span className="label">Email :</span>
          <input
            type="email"
            value={email}
            onChange={(e) => setEmail(e.target.value)}
            placeholder="Enter email"
          />
        </div>
        <button type="submit" style={{ padding: '6px 12px',
backgroundColor: '#28a745', color: '#fff', border: 'none',
borderRadius: '4px', cursor: 'pointer' }}>
          Submit
        </button>
      </form>
      {submitted && (
        <div style={{ marginTop: '16px', padding: '12px',
backgroundColor: '#d4edda', border: '1px solid #c3e6cb',
borderRadius: '4px' }}>
          <h3>Submitted Data:</h3>
          <p>Name: {submitted.name}</p>
          <p>Email: {submitted.email}</p>
        </div>
      )}
    </div>
  </div>
 );
};
export default UserForm;
```

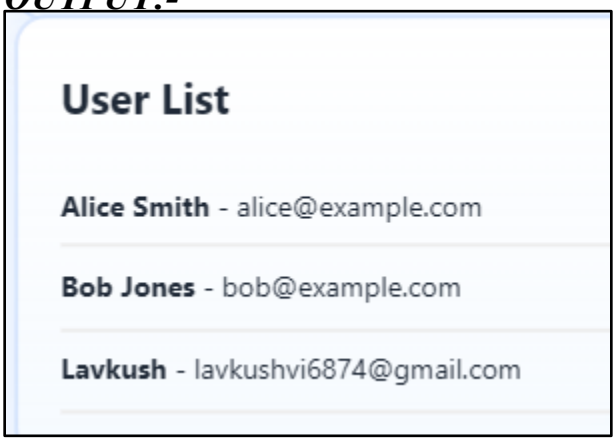
4.E). Display a list of items stored in the state using map().

SOURCE CODE:-

UserList.jsx

```
import React from 'react';
const UserList = ({ users }) => {
 return (
  <div className="card">
   <h2>User List</h2>
   <div className="info">
    <ul style={{ listStyle: 'none', padding: 0 }}>
     {users.map((user) => (
      <li key={user.id} style={{ padding: '8px 0',
borderBottom: '1px solid #eee' }}>
       <strong>{user.name}</strong> - {user.email}
      </li>
     ))}
    </ul>
   </div>
  </div>
 );
};
export default UserList;
```

OUTPUT:-

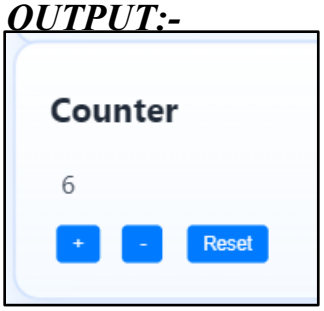


4.F). Build a small interactive component (e.g., counter or to-do list).

SOURCE CODE:-

```
import React, { useState } from 'react';
const Counter = () => {
 const [count, setCount] = useState(0);
 const buttonStyle = {
  padding: '6px 12px',
  backgroundColor: '#007bff',
  color: '#fff',
  border: 'none',
  borderRadius: '4px',
  cursor: 'pointer',
  margin: '0 4px',
 };
 return (
  <div className="card">
   <h2>Counter</h2>
   <div className="info">
    <span className="value">{count}</span>
```

OUTPUT:-




```
    <div style={{ display: 'flex', gap: '8px', marginTop:
'12px' }}>
     <button onClick={() => setCount(count + 1)}
style={buttonStyle}>
      +
     </button>
     <button onClick={() => setCount(count - 1)}
style={buttonStyle}>
```

```
          -
        </button>
        <button onClick={() => setCount(0)}
style={buttonStyle}>
          Reset
        </button>
```

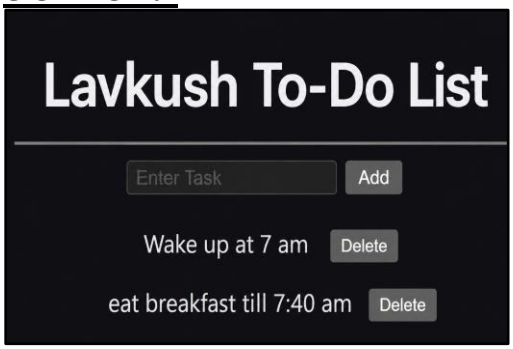
Q)To do List:-

SOURCE CODE:-

App.jsx:-

```
import { useState } from "react";
function App() {
  const [task, setTask] = useState("");
  const [tasks, setTasks] = useState([]);
  // Add Task
  const addTask = () => {
    if (task.trim() !== "") {
      setTasks([...tasks, task]);
      setTask("");
    }
  };
  // Delete Task
  const deleteTask = (index) => {
    const newTasks = tasks.filter((_, i) => i !== index);
    setTasks(newTasks);
  };
  return (
    <div>
      <h1>To-Do List</h1>
      <hr />
      <input
        type="text"
        placeholder="Enter Task"
        value={task}
        onChange={(e) => setTask(e.target.value)}
      />
      <button onClick={addTask}>Add</button>{" "}
      <ul>
        {tasks.map((item, index) => (
          <li key={index}>
            {item}
             
            <button onClick={() => deleteTask(index)}>
Delete</button>
          </li>
        ))}
      </ul>
    </div>
  );
}
export default App;
```

OUTPUT:-




```
      </div>
    </div>
  </div>
 );
};
export default Counter
```